Bio-inspired engineering: quieter, slipperier vehicles

Topic: noise-reduced and aerodynamically improved vehicles and aircraft

Team name and presenters here

What we will cover

- Why copy nature?
- Three core cases: kingfisher, owl, humpback whale
- Albatross soaring and bird-inspired winglets
- Live simulation, challenges, takeaways

Evolution is 3.8 billion years of R&D

- Natural shapes survive only if they work
- Biomimicry: observe, abstract the principle, then engineer it
- Popularised by Janine Benyus (1997)

Aerodynamics in 60 seconds

- Drag = pressure drag + skin-friction drag
- Streamlined shapes keep flow attached
- Separated flow makes a wake, which means more drag and more noise

Where vehicle noise comes from

- Turbulence over surfaces and edges
- Vortex shedding at trailing edges
- Pressure waves, such as a train entering a tunnel
- Engine, fan and propeller tones

The biomimicry design loop

- Find a problem
- Find an organism that solved it
- Extract the physical principle
- Prototype, wind-tunnel test, iterate

Kingfisher and the Shinkansen

Case 1 of 3

The problem: tunnel boom

- A fast train pushes a compression wave ahead of it in a tunnel
- The wave exits as a loud boom, a "micro-pressure wave"
- Complaints near tunnel exits limited speeds on Japan’s Sanyo line

Nature’s trick: the beak

- Kingfishers dive from air into denser water
- A long tapered beak makes density change gradually
- Result: a small splash and little pressure shock

The design process

- Engineer and bird-watcher Eiji Nakatsu led the JR West team
- Tested nose shapes in wind tunnels and pressure-wave rigs
- Chose a long, narrow, beak-like nose

Reported results

- 500 Series (1997) was among the world’s fastest trains in service
- Tunnel boom reduced; less power use at speed reported
- Verify exact figures in the citations tab

Lessons and where it’s used

- Gradual cross-section change calms pressure waves
- Later Shinkansen noses follow the same idea
- Trade-off: long noses cut passenger space

The owl and silent flight

Case 2 of 3

The problem: aerodynamic noise

- Turbulence hitting edges makes broadband noise
- Hard to remove with engine fixes alone
- Owls hunt prey they can’t afford to alarm

Three owl features

- Serrated leading edge
- Soft trailing-edge fringe
- Velvety, flexible surface fibres

Leading-edge comb

- Splits incoming flow into small, orderly vortices
- Delays transition to chaotic turbulence
- Copied as serrations on fan and propeller blades

Trailing-edge fringe

- Smooths the meeting of upper and lower flow
- Reduces sound radiated at the edge
- Copied as serrated trailing edges on wind turbines

Velvet surface

- Fine fibres damp pressure fluctuations on the wing
- Reduces roughness and vortex noise
- Tested as soft coatings and porous surface treatments

Applications

- Wind turbine blade serrations
- Quiet fans, drones and propellers
- Research on airliner wings, slats and landing gear

The humpback whale flipper

Case 3 of 3

Nature’s trick: tubercles

- Humpback flippers have bumps on the leading edge
- Counterintuitive: a wavy edge outperforms a smooth one
- Whales turn tightly while hunting

How tubercles work

- Each bump channels flow into a pair of counter-rotating vortices
- Those vortices re-energise the boundary layer
- Flow stays attached at higher angles

Reported results

- Wind-tunnel tests: later stall, higher peak lift, less drag past stall
- Stall-delay benefits seen on models; effect depends on shape
- Check numbers in citations before quoting

Applications

- Wind turbine blades (Whale Power)
- Fans and ventilation
- Research on aircraft wings, tails and control surfaces

Albatross: the glide champion

- Wingspans over 3 m on the largest species
- Long, thin wings give high lift for low induced drag
- Can cover huge distances with very few wingbeats

Dynamic soaring

- Wind is slower near the waves and faster above
- Climbing into fast air and diving back gains energy
- Studied for drones that harvest wind instead of fuel

Winglets: bird wingtips

- Soaring birds spread their tip feathers
- Winglets reduce induced drag from tip vortices
- Standard on many airliners today

Side by side

- Kingfisher: pressure waves and drag
- Owl: turbulence noise
- Humpback: stall delay and noise
- Albatross: efficient gliding

Challenges

- Manufacturing complex surfaces
- Scaling from animals to full-size vehicles
- Durability: ice, dirt and wear
- Proving gains on real aircraft

Future directions

- 3D-printed bio-structures
- Morphing and flexible wings
- AI-assisted bio-search
- Combined noise and drag designs

Key takeaways

- Nature solves physics problems with shape
- Three animals, three flow-control tricks
- Engineering turns insight into testable designs

Questions?

Thank you

Try the Simulation tab